

A GAP package for local action diagrams

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Why local action diagrams?

Local action diagrams are graph-based combinatorial structures. Their isomorphism classes are in one-to-one correspondence with isomorphism classes of actions of groups on trees with Tits' independence property (P), so-called (P)-closed actions, see [RS26] and [RS23].

Theorem (Reid–Smith '20)

$$\left\{ \begin{array}{l} \text{Pairs } (G, T) \\ G \leq \text{Aut}(T) \text{ is } (P)\text{-closed} \end{array} \right\} / \cong \xrightarrow{1:1} \left\{ \begin{array}{l} \text{local action} \\ \text{diagrams} \end{array} \right\} / \cong$$

What is a local action diagram?

Definition (Reid–Smith '20)

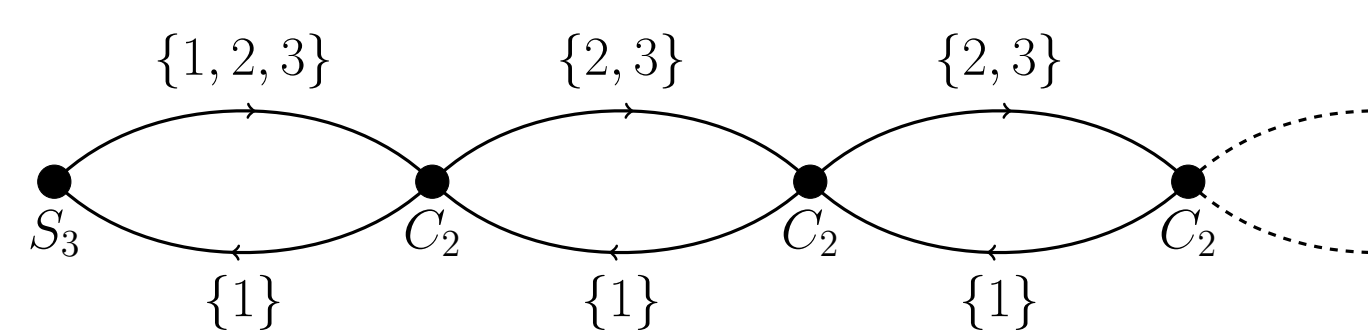
A **local action diagram** Δ is a triple $(\Gamma, (X_a)_{a \in A\Gamma}, (G(v))_{v \in V\Gamma})$ consisting of

- a connected graph Γ ,
- pairwise disjoint, non-empty sets X_a ($a \in A\Gamma$), and
- closed subgroups $G(v) \leq \text{Sym}(X_v)$ ($v \in V\Gamma$), where $X_v := \cup_{a \in o^{-1}(v)} X_a$, such that the sets X_a ($a \in o^{-1}(v)$) are precisely the orbits of $G(v)$.

The X_a are **colour sets**, its elements **colours**, and the $G(v)$ **local actions**.

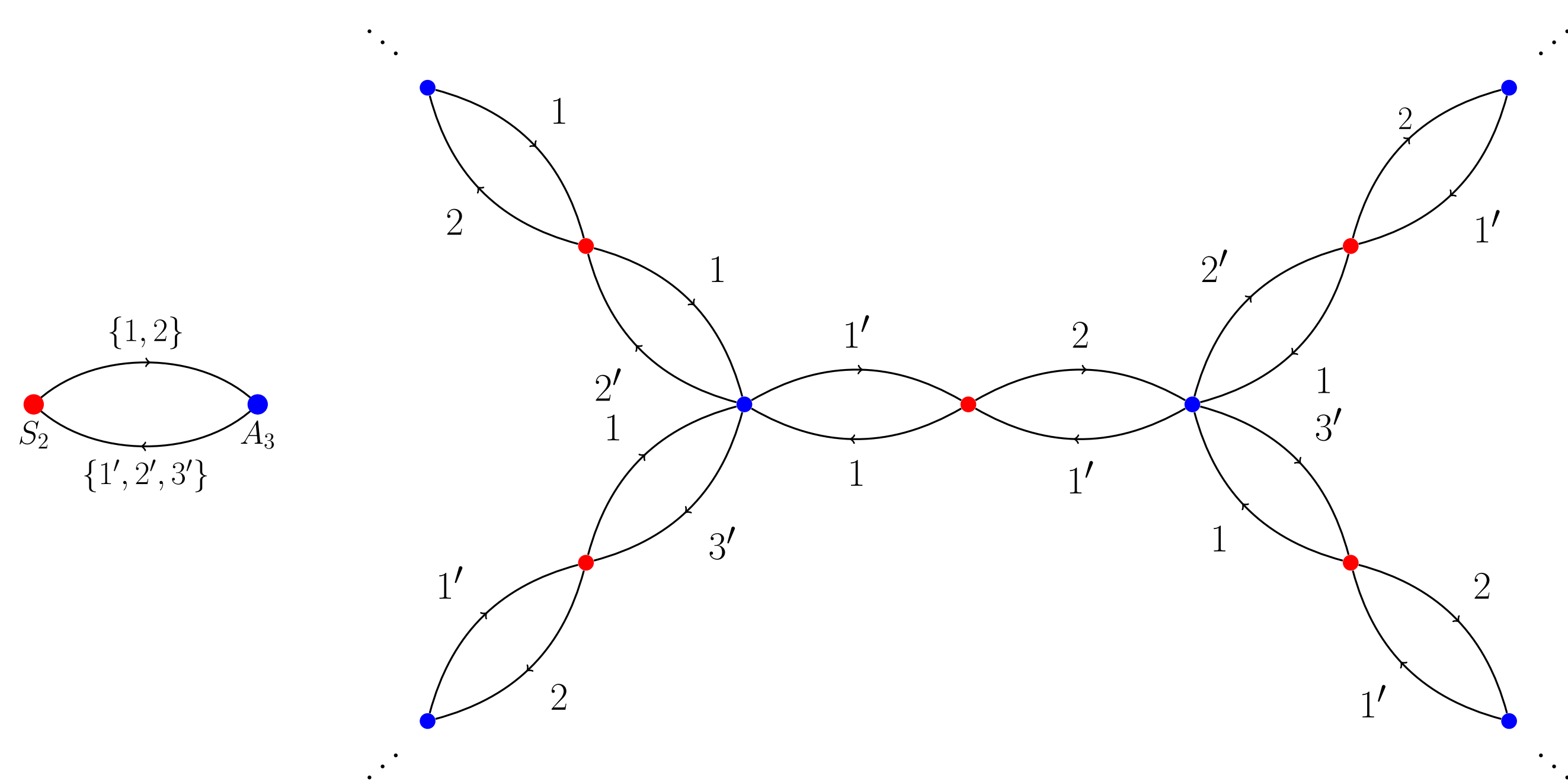
From groups to diagrams

Given a group G acting on a tree T , a local action diagram $\Delta = (\Gamma, (X_a)_a, (G(v)))$ arises from the quotient graph $\Gamma := G \backslash T$ by labelling it with the local actions of G acting on T , i.e., the permutation groups induced by vertex stabilisers on neighbours, and their orbits. For example, the local action diagram associated to a vertex stabiliser in a 3-regular tree looks as follows.



From diagrams to groups

Starting from a local action diagram $\Delta = (\Gamma, (X_a)_a, (G(v)))$, one first constructs a so-called Δ -tree $\mathbf{T} = (T, \pi, \mathcal{L})$, essentially the universal cover of the diagram with colour sets taken into account. It consists of a tree T together with a projection π onto Γ and an arc-labelling $\mathcal{L}$.



Given $g \in \text{Aut}_\pi(T)$ and $v \in VT$, a *local action* $\sigma_{\mathcal{L}}(g, v) \in \text{Sym}(X_v)$ is induced by g at v .

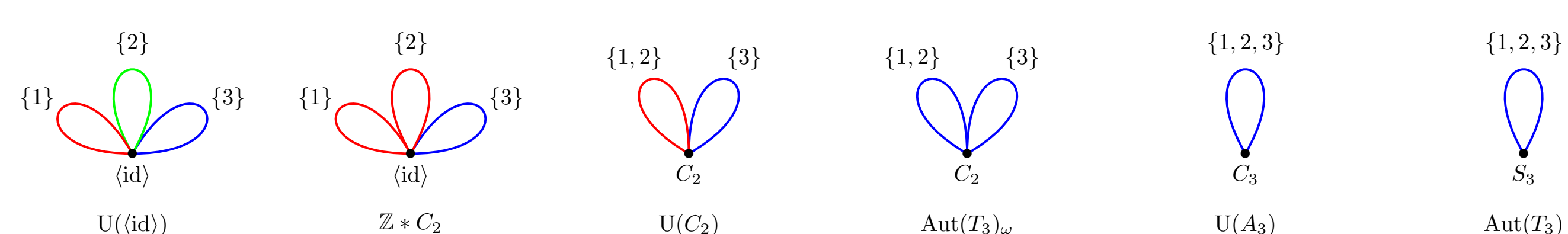
Definition (Reid–Smith '20)

Let $\Delta = (\Gamma, (X_a)_a, (G(v)))$ be a local action diagram and $\mathbf{T} = (T, \pi, \mathcal{L})$ be a Δ -tree. The **universal group of $\mathbf{T}$ with respect to the local actions** $(G(v))_v$ is

$$\mathbf{U}(\mathbf{T}, (G(v))) := \{g \in \text{Aut}_\pi(T) \mid \forall v \in VT : \sigma_{\mathcal{L}, v}(g) \in G(\pi(v)) \leq \text{Sym}(X_{\pi(v)})\}.$$

More examples

The following is a complete list of conjugacy classes of vertex-transitive (P)-closed subgroups of $\text{Aut}(T_3)$ and their local action diagrams. Each loop is inverse to the loop of the same colour.



The GAP package

A GAP package pertaining to local action diagrams is work in progress, see [CT26]. Thanks to the theorem, properties of the acting group correspond to properties of the local action diagram. Establishing this dictionary is ongoing theoretical work. It includes the following.

- local compactness, compact generation, simplicity [RS26]
- group action type, discreteness [CT24]
- scale function, modular function [CFT26]

The package itself provides, in the finite case, a data structure for the graphs underlying local action diagrams as well as the diagrams themselves. It further provides functions to test for the most of the above properties, and enables isomorphism checking for both graphs and diagrams, resulting in a library similar to the transitive groups library that a user may query.

Code examples

```
# The graphs underlying local action diagrams may be created from scratch, or the library.
gap> RSGraphByAdjacencyList([[1,1],[1,2],[2,1],[2,2],[2,2],[2,3]],(2,3)(4,5));
<RSGraph with 2 vertices and 5 arcs>
# Printing the underlying data of this graph.
gap> Print(last);
Vertices = { 1, 2 }
Arcs = {
  1 = ( origin = 1, terminus = 1, inverse = 1 )
  2 = ( origin = 1, terminus = 2, inverse = 3 )
  3 = ( origin = 2, terminus = 1, inverse = 2 )
  4 = ( origin = 2, terminus = 2, inverse = 5 )
  5 = ( origin = 2, terminus = 2, inverse = 4 )
}
Reverse Map = (2,3)(4,5)
# Local action diagrams may be created from scratch, special functions, or the library
gap> LocalActionDiagramFromBurgerMozesUniversalGroup(AlternatingGroup(4));
<U(AlternatingGroup( [ 1 .. 4 ] )) (as a Local Action Diagram)>
# Loading all local action diagrams of groups acting on T3 with 1 vertex orbit
gap> LocalActionDiagramFromLibrary(3,1);
[ <U(A3) (as a Local Action Diagram)>,
  <Aut(T3) (as a Local Action Diagram)>,
  <U(C2) (as a Local Action Diagram)>,
  <Aut(T3)_omega (as a Local Action Diagram)>,
  <U(1) (as a Local Action Diagram)>,
  <Z ast C2 (as a Local Action Diagram)> ]
# Printing the underlying data of Aut(T3)_omega
gap> Print(last[4]);
Vertices = { 1 }
Arcs = {
  1 = ( origin = 1, terminus = 1, inverse = 2 )
  2 = ( origin = 1, terminus = 1, inverse = 1 )
}
Reverse Map = (1,2)
Vertex Labels = {
  1 = Group( [ (2,3) ] )
}
Arc Labels = {
  1 = [ 1 ]
  2 = [ 2, 3 ]
}
# Determining the number of (P)-closed groups acting on T5 with 2 vertex orbits...
gap> lad52:=LocalActionDiagramFromLibrary(5,2);
gap> Size(lad52);
1449
# ...as well as how many of them are unimodular and discrete respectively...
gap> Size(Filtered(lad52,LocalActionDiagramIsUnimodular));
880
gap> Size(Filtered(lad52,LocalActionDiagramIsDiscrete));
18
# ...and the split between different group action types
gap> List(lad52,Delta->LocalActionDiagramGroupType(Delta));
gap> Collected(last);
[ [ "Focal", 15 ], [ "General", 1434 ] ]
```

References

- [CFT26] M. Chijoff, M. Ferov, and S. Tournier, *The of (P)-closed groups acting on trees*, arXiv preprint 2606.27998 (2026).
- [CT24] M. Chijoff and S. Tournier, *Discrete (P)-closed groups acting on trees*, arXiv preprint 2409.13240 (2024).
- [CT26] M. Chijoff and S. Tournier, *A GAP package for local action diagrams*, 2026, <https://github.com/marcusc31415/LocalActionDiagrams>.
- [RS23] C. Reid and S. Smith, *An introduction to the local-to-global behaviour of groups acting on trees and the theory of local action diagrams*, arXiv preprint 2309.05065 (2023).
- [RS26] ———, *Groups acting on trees with Tits' independence property (P): With an appendix by Stephan Tournier*, Mathematische Annalen **395** (2026), no. 3, 53.